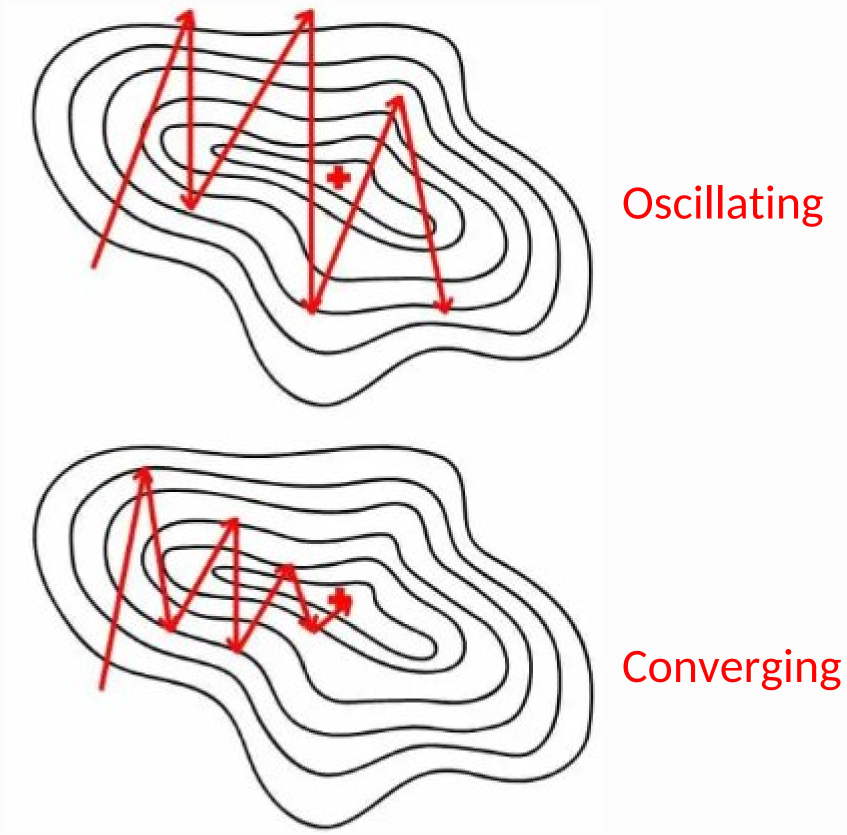
Learning Rate Schedulers

A concise overview of how learning rate schedulers improve model training

Why Use Learning Rate Schedulers

Fixed learning rates often lead to poor convergence or oscillation during training. Schedulers dynamically adjust the learning rate throughout the training process, providing adaptive control over optimization speed.

This dynamic adjustment improves convergence speed, enhances model generalization, and promotes training stability across different phases of the learning process.



Common Learning Rate Schedulers

Step Decay

Reduces learning rate by a fixed factor every few epochs, creating discrete drops in the learning rate schedule.

Exponential Decay

Learning rate decreases exponentially with time, providing smooth continuous reduction throughout training.

Cosine Annealing

Smoothly reduces learning rate following a cosine curve, avoiding abrupt changes for better convergence.

Cyclic / OneCycle

Periodically increases and decreases learning rate to help escape local minima and improve exploration.

We will cover the ones in Green

Cosine Annealing: Intuition

01

Inspired by Simulated Annealing

The approach mimics the physical process of 'cooling' the optimizer over time, gradually reducing exploration.

02

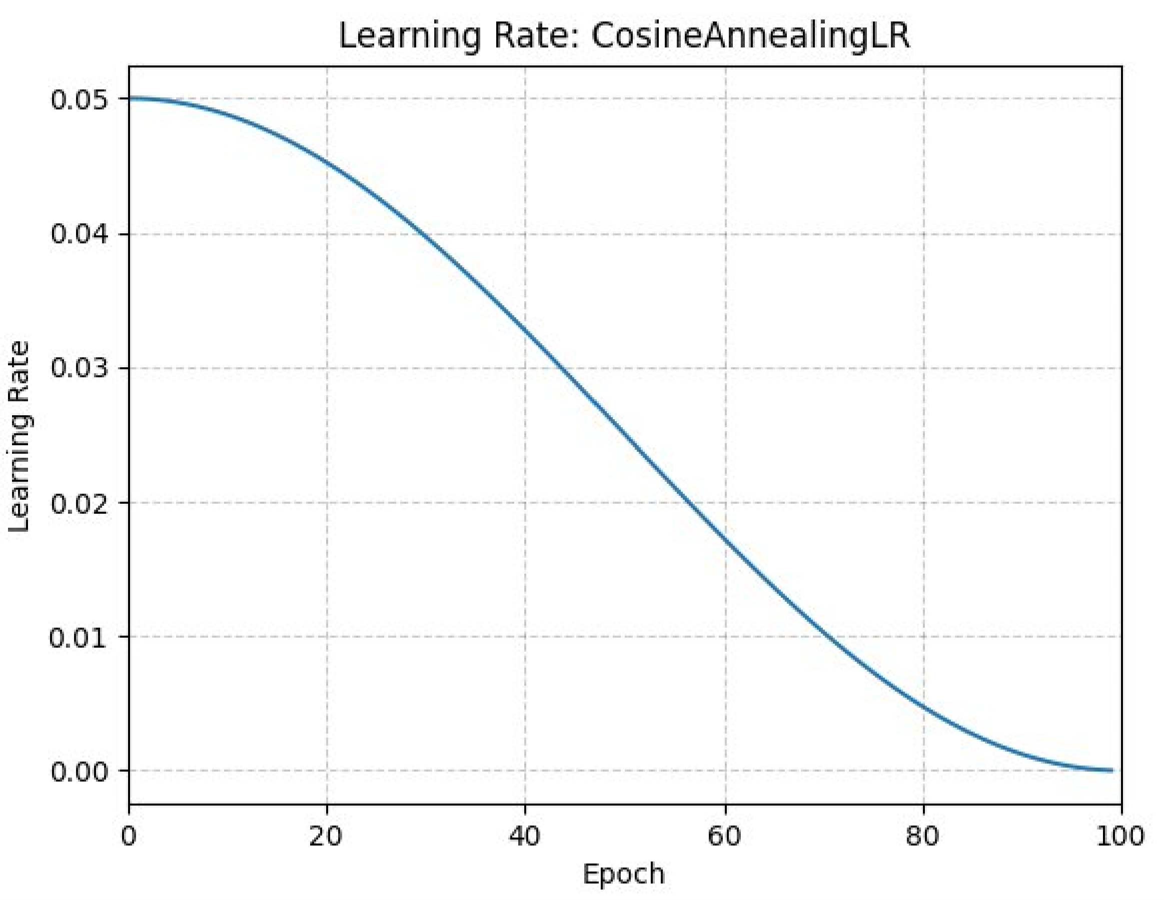
Smooth Cosine Curve

Starts with a high learning rate, then gradually decreases following a cosine curve, avoiding abrupt changes.

03

Balanced Training Strategy

Encourages broad exploration early in training, then transitions to fine-tuning for optimal convergence.



Cosine Annealing Formula

The learning rate at **epoch *t*** is calculated using the following formula:

$$LR(t) = \eta_{min} + \frac{\eta_{max} - \eta_{min}}{2} \times (1 + \cos(\frac{\pi \times t}{T_{max}}))$$

ηmax

Initial (maximum) learning rate at the start of training

ηmin

Final (minimum) learning rate at the end of the schedule

Tmax

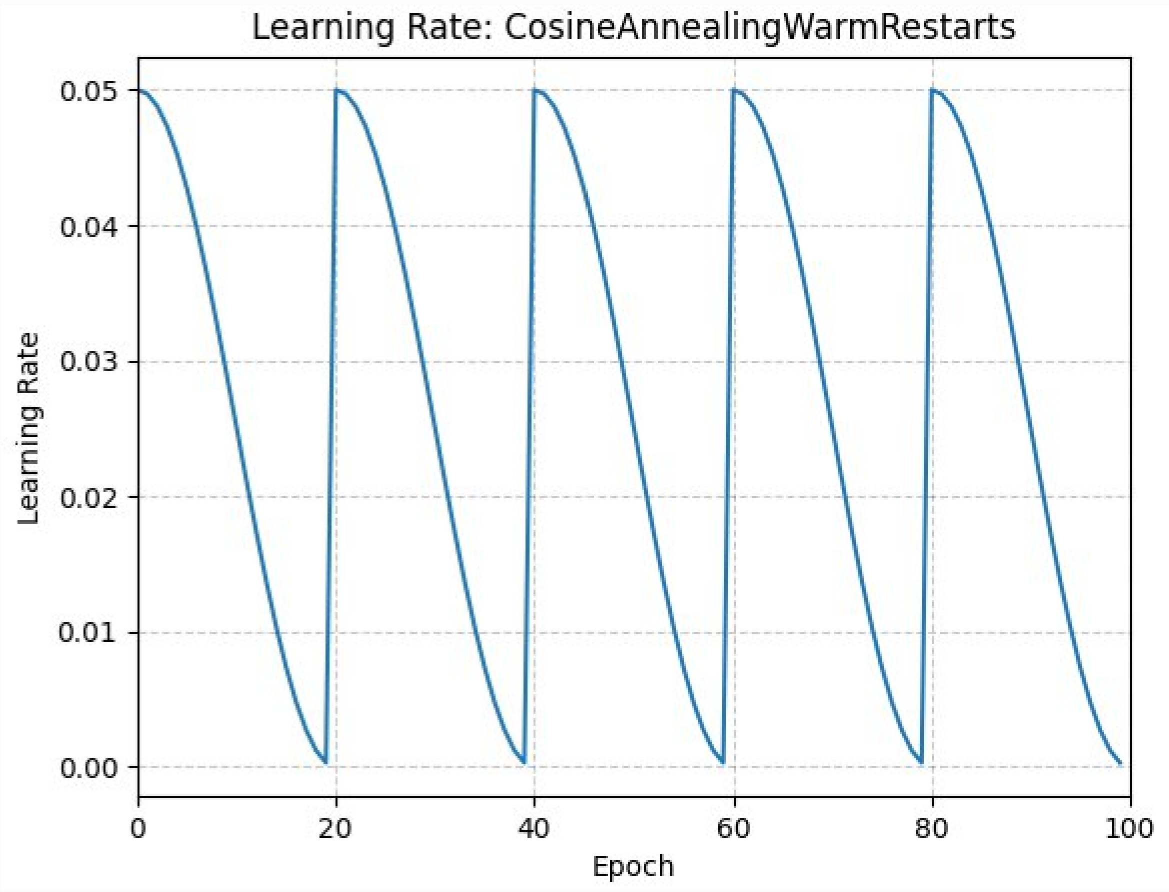
Number of epochs before restart

Cosine Annealing with Warm Restarts (SGDR)

Introduced in **'SGDR: Stochastic Gradient Descent with Warm Restarts'** by Loshchilov & Hutter (2016), this approach extends basic cosine annealing with periodic restarts.

After Tmax iterations, the learning rate resets to ηmax and decays again. Each restart helps the model escape local minima and explore different regions of the loss landscape.

PyTorch Implementation: Use `torch.optim.lr_scheduler.CosineAnnealingLR` or `CosineAnnealingWarmRestarts` for easy integration.



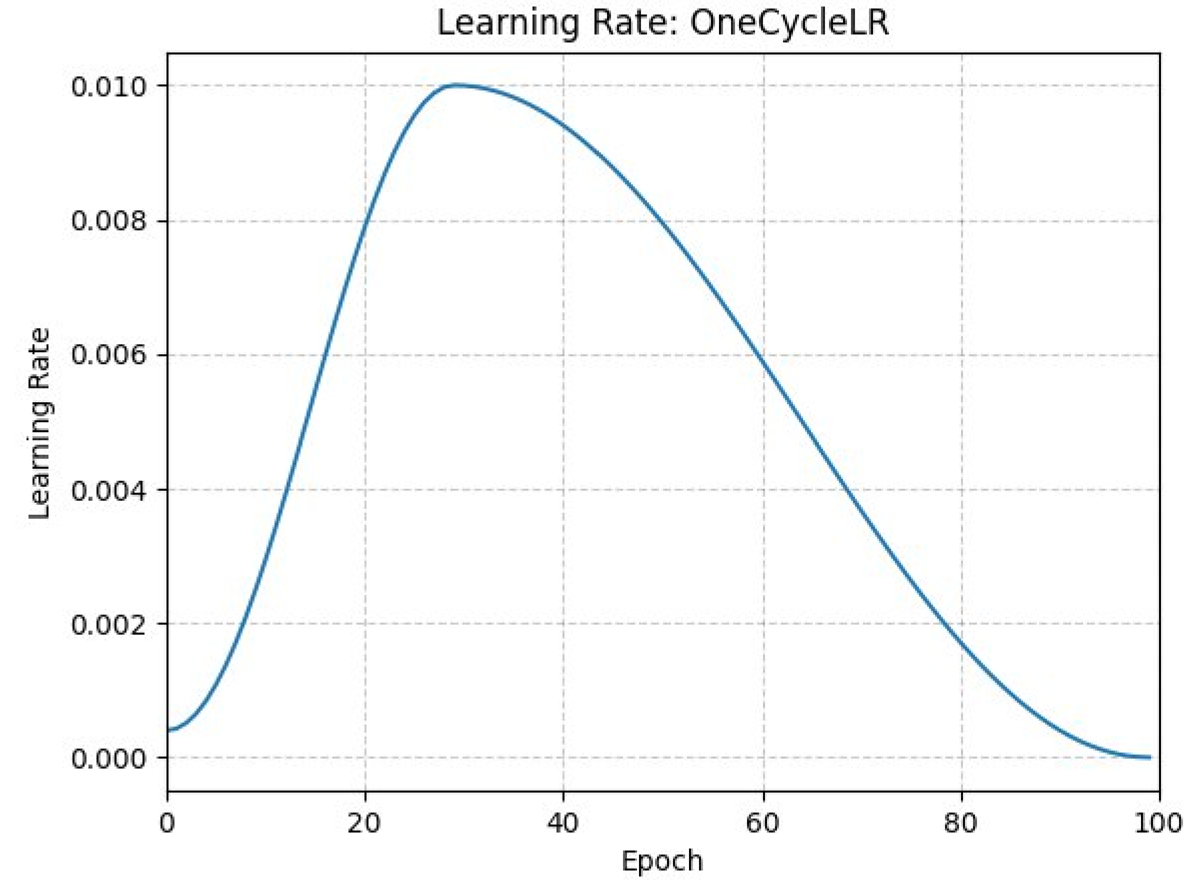
One Cycle

Introduced in '[Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates]', The 1cycle policy anneals the learning rate from an initial learning rate to some maximum learning rate and then from that maximum learning rate to some minimum learning rate much lower than the initial learning rate.Creats a single cycle that balances exploration and convergence throughout training.

Pct_start defines the percentage of the cycle increasing the learning rate (default= .3)

See docs for other params

PyTorch Implementation: Use `torch.optim.lr_scheduler.OneCycleLR` or for easy integration.



Why are we talking about this?

LLMs use these schedulers, both during **pre-training** and **fine-tuning**. They help:

- **Stabilize training** (warmup prevents early divergence)
- **Escape sharp minima** (cosine/step decay)
- **Avoid over-training late** (schedule forces gradual improvement)
- **Improve generalization**

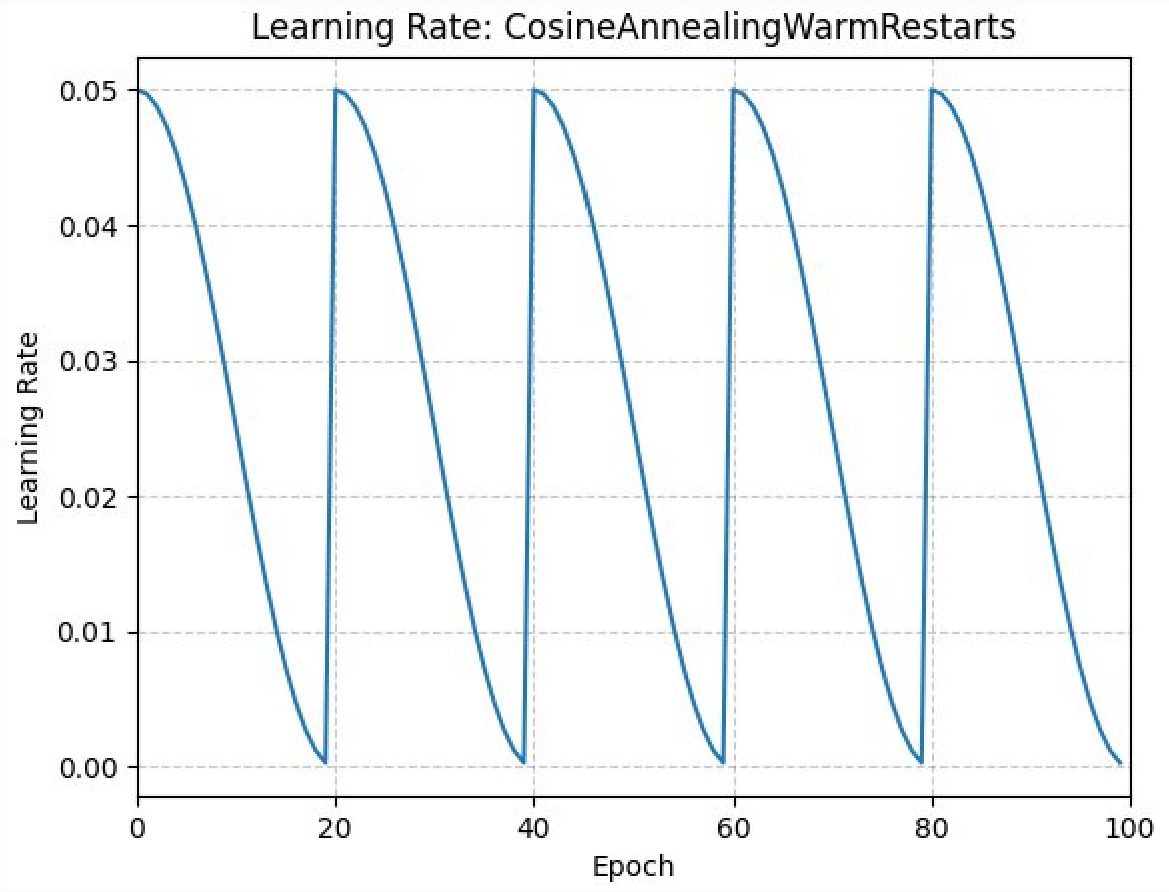
The **warmup** phase is extremely common for transformers:

Typically 2–5% of steps are warmup, then LR decays.

Mental model

Warmup lets the model **learn to walk before running**

Decay prevents it from **overshooting optimal minima**



Important: Do NOT Use Early Stopping with Schedulers

Critical Warning: Learning rate schedulers like Cosine Annealing and One Cycle are designed to control the full training trajectory. Stopping early interrupts their intended schedule and prevents the model from reaching its optimal point.

1. **Complete the Cycle**

   Let the scheduler complete its planned cycle or epochs as designed.

2. **Adjust Parameters**

   Use validation performance to adjust scheduler parameters, not to stop training.

3. **Reduce Max Epochs**

   If overfitting is a concern, consider reducing max epochs instead of early stopping.

Key Takeaway: Learning rate schedulers adapt training speed dynamically. Cosine annealing provides smooth, non-linear decay that improves convergence and model generalization. Warm restarts enhance performance by reintroducing exploration phases throughout training.